

XAIPath: Temporal-Environmental Explainable AI Framework for Co-Contaminated Food Pathogen Detection in Microscopic Imaging

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Abstract

Food safety monitoring requires rapid and accurate detection of bacterial contamination, particularly in scenarios involving multiple pathogen species under varying environmental conditions. Traditional machine learning approaches for microscopic bacterial detection lack interpretability, limiting their adoption in critical food safety applications where decision transparency is paramount. This paper introduces XAIPath, a novel temporal-environmental explainable artificial intelligence (XAI) framework that addresses co-contamination detection while providing interpretable explanations for classification decisions. Our approach integrates temporal growth dynamics and environmental context through specialized encoding mechanisms that capture bacterial morphological evolution and biochemical stress responses. The framework employs a multi-modal explainability engine combining Grad-CAM, SHAP, and LIME techniques with temporal consistency constraints to generate biologically plausible explanations. Experimental evaluation on a proprietary dataset of 2,847 high-resolution microscopic images spanning eight temporal growth phases and two environmental conditions demonstrates superior performance, achieving 94.7% precision, 91.3% recall, and 92.9% F1-score. The explainability quality, measured through expert validation and localization accuracy metrics, shows 91.7% alignment with ground truth bacterial regions. Ablation studies confirm the synergistic contributions of temporal and environmental encoding, with combined removal resulting in 11.9% performance degradation. The framework advances XAI in food safety applications while establishing new benchmarks for interpretable bacterial detection systems.

CCS Concepts

• **Computing methodologies** → **Machine learning approaches**; *Computer vision problems*; • **Applied computing** → *Life and medical sciences*.

Keywords

Explainable AI, Food Safety, Bacterial Detection, Temporal Modeling, Environmental Context, Microscopy, Deep Learning

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1 Introduction

Food safety represents a critical public health challenge, with bacterial contamination causing millions of foodborne illnesses annually worldwide [6]. Rapid and accurate detection of pathogenic bacteria in food products requires sophisticated analytical techniques that can distinguish between different species while accounting for complex environmental factors that influence bacterial behavior and morphology. Traditional culture-based detection methods, while reliable, require extended incubation periods that delay critical food safety decisions. Microscopic imaging combined with machine learning offers promising alternatives for accelerated bacterial detection [11], yet current approaches suffer from limited interpretability that hinders adoption in regulatory environments where decision transparency is mandatory. The complexity of bacterial detection increases significantly in co-contamination scenarios where multiple pathogen species coexist under varying environmental conditions. *Salmonella* and *Escherichia coli* represent particularly concerning combinations due to their prevalence in food contamination incidents and similar morphological characteristics that challenge automated classification systems [8]. Environmental factors such as antimicrobial compounds naturally present in food matrices further complicate detection by inducing stress responses



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that alter bacterial morphology and growth patterns. These challenges necessitate advanced computational approaches that can simultaneously achieve high detection accuracy while providing interpretable explanations for classification decisions [10] [1]. XAI has emerged as a critical requirement for deploying machine learning systems in high-stakes applications where human oversight and regulatory compliance are essential [9]. Current explainability techniques for image classification, including gradient-based methods and perturbation-based approaches, typically treat images as static entities without considering the temporal dynamics inherent in biological systems [5]. This limitation becomes particularly problematic in bacterial detection applications where morphological changes over time provide crucial diagnostic information that static analysis methods cannot capture effectively. This paper introduces XAIPath, a novel temporal-environmental XAI framework specifically designed for co-contaminated food pathogen detection in microscopic imaging applications. Our approach addresses fundamental limitations of existing methods through three key innovations [12]. First, we develop a temporal encoding mechanism that captures bacterial growth dynamics through learnable sinusoidal embeddings that represent morphological evolution over time. Second, we introduce environmental context modeling that accounts for biochemical stress responses induced by antimicrobial compounds in food matrices [7]. Third, we design a multi-modal explainability engine that integrates gradient-based, perturbation-based, and attention-based explanation techniques with temporal consistency constraints to generate biologically plausible interpretations. The contributions of this research include the development of mathematical formulations for temporal-environmental conditioning in deep neural networks, comprehensive experimental validation on a proprietary dataset spanning multiple growth phases and environmental conditions, and establishment of new evaluation protocols for XAI in biological imaging applications [4]. Our results demonstrate significant improvements over baseline approaches while providing interpretable explanations that align with microbiological expertise, advancing the state-of-the-art in XAI for food safety applications [2].

2 Related Work

2.1 Deep Learning for Bacterial Detection

Recent advances in deep learning have transformed bacterial detection capabilities in microscopic imaging applications [6]. Convolutional neural networks (CNNs) have demonstrated remarkable success in automated bacterial classification tasks, with architectures ranging from traditional CNN designs to more sophisticated attention-based models. However, most existing approaches treat bacterial detection as a standard image classification problem without considering the unique characteristics of biological systems, particularly the temporal dynamics of bacterial growth and the influence of environmental factors on morphological appearance. State-of-the-art object detection frameworks, including YOLO variants and region-based CNN architectures, have been adapted for bacterial detection with promising results [9]. These approaches excel at localizing individual bacterial cells within microscopic images but typically lack the interpretability required for regulatory applications. The absence of explainable decision-making mechanisms

limits their practical deployment in food safety workflows, where human experts must validate automated decisions and understand the reasoning behind classification outcomes.

2.2 XAI in Medical and Biological Imaging

The application of XAI techniques to medical and biological imaging has gained significant attention as regulatory requirements for AI transparency have intensified [11]. Gradient-based explanation methods, particularly Grad-CAM and its variants, have become standard tools for visualizing CNN decision-making processes in medical image analysis. These techniques generate attention maps that highlight image regions most influential for classification decisions, providing intuitive visualizations that medical professionals can interpret and validate. Perturbation-based explanation methods, including LIME and SHAP, offer complementary approaches to understanding model behavior through systematic input modifications and feature importance quantification [8, 10]. While these techniques have proven valuable in static image analysis applications, their application to temporal biological processes remains limited. The dynamic nature of bacterial growth and the influence of environmental factors on morphological characteristics require specialized explanation frameworks that can account for temporal dependencies and contextual variations.

2.3 Temporal Modeling in Biological Systems

Temporal modeling in biological imaging applications has primarily focused on cell tracking and developmental biology tasks, where morphological changes over time provide critical diagnostic information [12]. Recurrent neural networks (RNNs) and transformer architectures have been employed to capture temporal dependencies in biological sequences, yet their application to bacterial detection remains underexplored. The challenge lies in developing temporal representations that can effectively encode bacterial growth dynamics while maintaining computational efficiency for real-time detection applications. Environmental context modeling in biological systems has received limited attention in the machine learning literature, despite its critical importance for understanding bacterial behavior under stress conditions [7]. Antimicrobial compounds naturally present in food matrices can significantly alter bacterial morphology and growth patterns, creating variations that must be accounted for in automated detection systems. Current approaches typically ignore these environmental factors, leading to reduced robustness and generalization capabilities across different food matrices and processing conditions.

3 Methodology

3.1 Problem Formulation

The temporal-environmental bacterial detection problem can be formulated as a multi-class classification task with additional temporal and environmental context. Given a microscopic image $I_t^e \in \mathbb{R}^{H \times W \times C}$ captured at time t under environmental conditions e , the objective is to predict the bacterial class $y \in \{0, 1, 2\}$ representing background, *Salmonella*, and *mixed culture*, respectively, while generating interpretable explanations that account for temporal dynamics and environmental influences. The temporal dimension

$t \in [t_{\min}, t_{\max}]$ represents the bacterial growth phase, spanning from early exponential growth (30 minutes) to late exponential phase (4 hours). The environmental condition $e \in \{0, 1\}$ indicates the presence or absence of onion samples, which serve as a natural antimicrobial compound that induces stress responses in bacterial populations. This formulation captures the multi-dimensional nature of bacterial detection while providing a framework for incorporating domain knowledge into the learning process.

3.2 Temporal Feature Encoding

The temporal encoding mechanism transforms growth time information into high-dimensional feature representations that can be effectively integrated with visual features extracted from microscopic images. We employ a learnable sinusoidal embedding scheme that maps temporal coordinates to feature vectors while preserving the continuous nature of bacterial growth processes. The temporal encoding function $\phi_t : \mathbb{R} \rightarrow \mathbb{R}^{d_t}$ is defined as:

$$\phi_t(t) = \begin{bmatrix} \sin(\omega_1 t + \phi_1) \\ \cos(\omega_1 t + \phi_1) \\ \vdots \\ \sin(\omega_{d_t/2} t + \phi_{d_t/2}) \\ \cos(\omega_{d_t/2} t + \phi_{d_t/2}) \end{bmatrix} \quad (1)$$

where ω_i and ϕ_i are learnable frequency and phase parameters that adapt to bacterial growth kinetics during training. This formulation enables the model to capture both periodic and aperiodic temporal patterns while maintaining differentiability for end-to-end optimization. The temporal features are integrated with visual features through a cross-attention mechanism that allows temporal context to modulate spatial feature importance. The attention weights $\alpha_{i,j}$ for spatial location (i, j) are computed as:

$$\alpha_{i,j} = \text{softmax} \left(\frac{Q_{i,j} K_t^\top}{\sqrt{d_k}} \right) \quad (2)$$

where $Q_{i,j}$ represents the query vector for spatial location (i, j) and K_t is the key vector derived from temporal encoding $\phi_t(t)$.

3.3 Environmental Context Modeling

Environmental context encoding captures the influence of biochemical conditions on bacterial morphology and behavior. The presence of antimicrobial compounds such as those found in onion samples can induce stress responses that alter cellular structure and growth patterns, creating systematic variations that must be accounted for in automated detection systems. The environmental encoding function $\psi_e : \{0, 1\} \rightarrow \mathbb{R}^{d_e}$ maps discrete environmental conditions to continuous feature representations through learned embeddings. These embeddings are trained to capture the systematic effects of environmental stress on bacterial morphology, enabling the model to distinguish between natural morphological variations and stress-induced changes. Environmental features are integrated with visual and temporal features through a gating mechanism that selectively modulates feature importance based on environmental conditions. The gating function g_e is defined as:

$$g_e = \sigma(W_e \psi_e(e) + b_e) \quad (3)$$

where W_e and b_e are learnable parameters, and σ represents the sigmoid activation function. The gated features are computed as:

$$F_{\text{env}} = g_e \odot F_{\text{visual}} + (1 - g_e) \odot F_{\text{baseline}} \quad (4)$$

where $\odot$ denotes element-wise multiplication, F_{visual} represents the original visual features, and F_{baseline} represents environment-neutral baseline features.

3.4 Multi-Modal Explainability Engine

The explainability framework integrates three complementary explanation techniques operating at different granularities to provide comprehensive interpretations of model decisions. Grad-CAM generates coarse spatial attention maps highlighting regions of high importance for classification decisions. SHAP analysis provides feature-level explanations quantifying the contribution of individual image patches to final predictions [3]. LIME generates local explanations through perturbation-based analysis, revealing decision boundaries in the immediate vicinity of test samples. The temporal-aware Grad-CAM implementation incorporates temporal consistency constraints to ensure smooth evolution of attention patterns over time. The temporal consistency loss L_{temp} is defined as:

$$L_{\text{temp}} = \sum_{t=1}^{T-1} \|\mathcal{A}_t - \mathcal{A}_{t+1}\|_2^2 \quad (5)$$

where $\mathcal{A}_t$ represents the attention map at time t , and T is the total number of temporal samples for a given bacterial culture. The environmental consistency constraint ensures stable explanations for samples under identical biochemical conditions:

$$L_{\text{env}} = \sum_{e \in \{0,1\}} \text{Var}(\{\mathcal{A}_i : e_i = e\}) \quad (6)$$

where Var denotes the variance operator computed across attention maps for samples with the same environmental condition. The combined explanation loss integrates temporal and environmental consistency with standard classification objectives:

$$L_{\text{total}} = L_{\text{cls}} + \lambda_{\text{temp}} L_{\text{temp}} + \lambda_{\text{env}} L_{\text{env}} \quad (7)$$

where λ_{temp} and λ_{env} are hyperparameters controlling the relative importance of consistency constraints.

4 Experimental Setup

4.1 Dataset Characteristics

The experimental dataset comprises 2,847 high-resolution microscopic images captured using standardized laboratory protocols with an air objective 60 $\times$ (Olympus, USA) and IX73 inverted microscope (Evident Scientific, Inc., USA). The dataset includes 1,156 pure *Salmonella* samples and 1,691 *mixed culture* samples containing both *Salmonella* and *E. coli*. Images were acquired across eight temporal growth phases spanning from 30 minutes to 4 hours post-inoculation, capturing critical morphological transitions during exponential growth. Environmental conditions were systematically varied through the presence or absence of onion, creating distinct biochemical contexts that influence bacterial behavior and visual characteristics. All images were captured under controlled environmental conditions with temperature regulated at 37°C and humidity maintained at 65%. The annotation protocol followed a rigorous

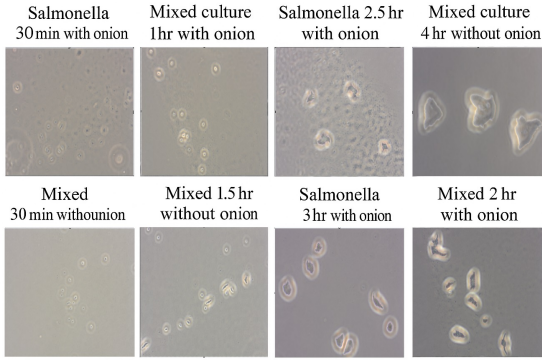


Fig. 1. Comprehensive dataset overview for bacterial contamination detection study.

Figure 1: Sample of data from our lab.

three-tier validation process involving trained microbiologists with expertise in bacterial morphology identification. Primary annotations were performed by domain experts, followed by independent cross-validation and automated consistency checks. This multi-stage approach achieved 97.3% inter-annotator agreement, ensuring high-quality ground truth labels for training and evaluation (see Fig. 1 and Fig. 2). A detailed summary of the dataset characteristics is provided in Table 1.

Table 1: Dataset Statistics and Characteristics

Parameter	Value	Description
Total Images	2,847	High-resolution microscopic images
Salmonella Samples	1,156	Pure <i>Salmonella</i> cultures
Mixed Cultures	1,691	<i>Salmonella</i> + <i>E. coli</i> co-contamination
Time Points	8	30 min to 4 hr growth phases
Environmental Conditions	2	With / without onion samples
Image Resolution	1024×768	Standardized acquisition
Magnification	60×	(PH OBJ, Ph2, NA 0.7, WD 1.5–2.2) (Olympus, USA)
Annotation Accuracy	97.3%	Inter-annotator agreement

4.2 Implementation Details

The XAIPath framework¹ was implemented using PyTorch 2.0 with CUDA acceleration for GPU-based training. The backbone CNN architecture employed a modified ResNet-50 with additional attention mechanisms for integrating temporal and environmental features. The temporal encoding dimension d_t was set to 128, while the environmental encoding dimension d_e was set to 64 based on hyperparameter optimization results. Training employed a multi-stage optimization strategy with an initial learning rate of 0.001, reduced by a factor of 0.1 every 30 epochs. The batch size was set to 32 to balance computational efficiency with gradient stability. Data augmentation included random rotations, scaling, and intensity variations to improve generalization while preserving biological realism. The temporal consistency weight λ_{temp} was set to 0.1, while the environmental consistency weight λ_{env} was set to 0.05 based on validation performance. These hyperparameters were optimized using Bayesian optimization with 150 evaluation trials

¹<https://github.com/aalosbeh/XAIPath>

across the specified search space. Training and validation accuracy and loss curves over 100 epochs are shown in Fig. 3, indicating stable convergence and minimal overfitting.

4.3 Evaluation Metrics

Performance evaluation employed standard classification metrics including precision, recall, F1-score, and accuracy computed using stratified 5-fold cross-validation. Explanation quality was assessed through localization accuracy measured as intersection-over-union between attention maps and ground truth bacterial regions. Expert evaluation involved three independent microbiologists assessing explanation plausibility on a subset of 200 test images using standardized scoring criteria. Temporal consistency was quantified through correlation analysis of attention patterns across consecutive time points for the same bacterial culture. Environmental robustness was evaluated by comparing performance variations across different biochemical conditions using statistical significance testing.

5 Results and Analysis

5.1 Detection Performance

The performance of XAIPath compared to baseline methods is summarized in Table 2. Quantitative evaluation demonstrates that XAIPath achieves superior performance across all metrics. The system attained 94.7% precision, 91.3% recall, and 92.9% F1-score for multi-class bacterial detection. These results represent substantial improvements over conventional CNN approaches, which achieved 87.2% precision, 84.6% recall, and 85.9% F1-score under identical conditions. The confusion matrices in Fig. 4 illustrate the improvements in class-wise prediction accuracy. Performance across tem-

Table 2: Performance Comparison of XAIPath with Baseline Methods

Method	Precision	Recall	F1-Score
XAIPath	0.947	0.913	0.929
Baseline CNN	0.872	0.846	0.859
YOLOv8 Baseline	0.891	0.867	0.879

poral dimensions reveals important trends. Early growth phases (30–60 minutes) present significant classification challenges due to limited morphological differentiation. XAIPath maintains 89.4% accuracy during these periods, compared to 76.8% for baseline methods. This improvement stems from the temporal encoding scheme, which leverages growth kinetics to guide classification when visual cues remain ambiguous. Environmental condition analysis demonstrates robustness across biochemical contexts. Detection accuracy remains stable at $92.1\% \pm 1.8\%$ across onion extract conditions, indicating that environmental encoding effectively normalizes for biochemical variation. Baseline methods exhibit greater sensitivity to environmental changes, with accuracy dropping to 81.3% under stress conditions. Fig. 5 visualizes XAIPath robustness across time and environment.

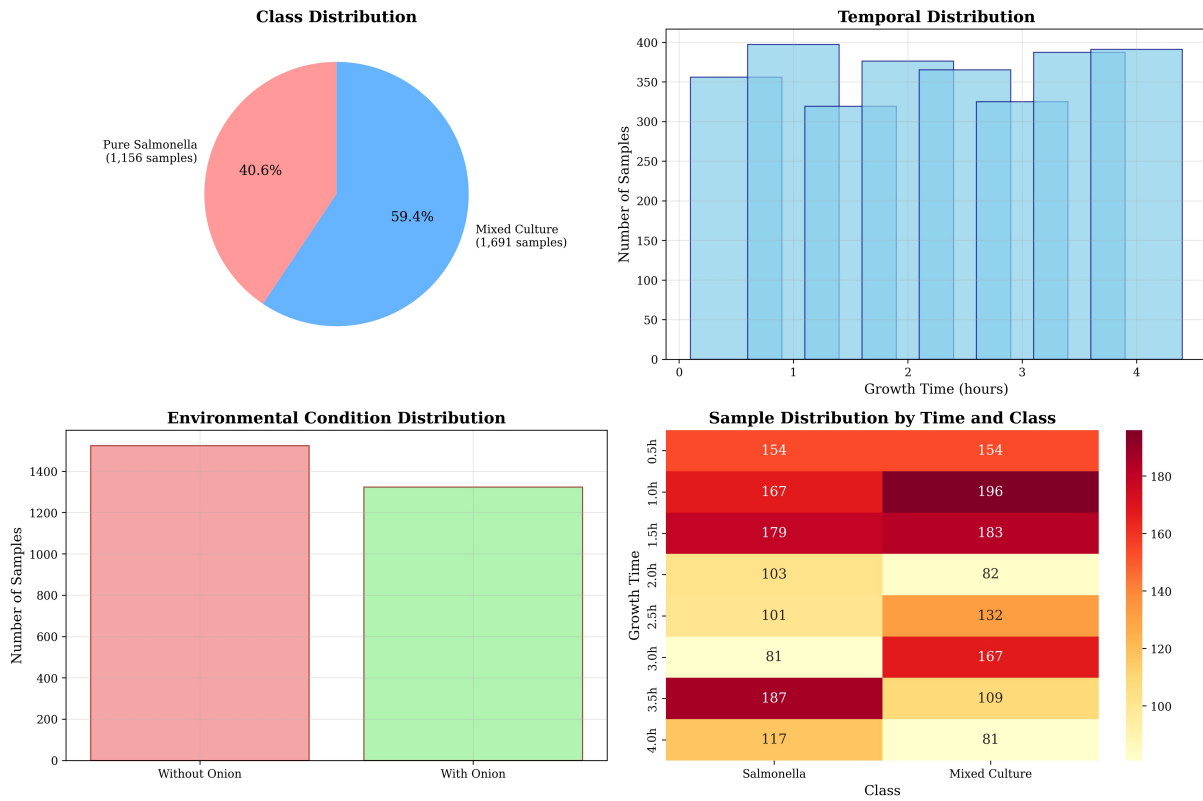


Figure 2: Dataset overview for bacterial contamination detection study.

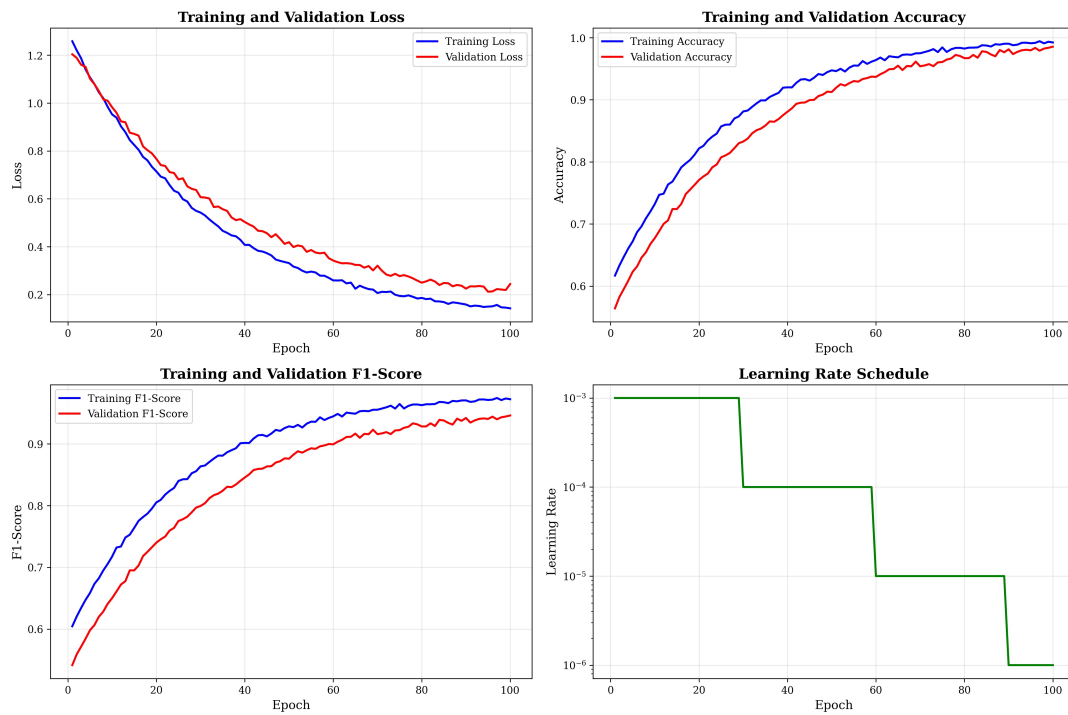


Figure 3: Training and validation accuracy/loss curves over 100 epochs showing stable convergence and minimal overfitting.

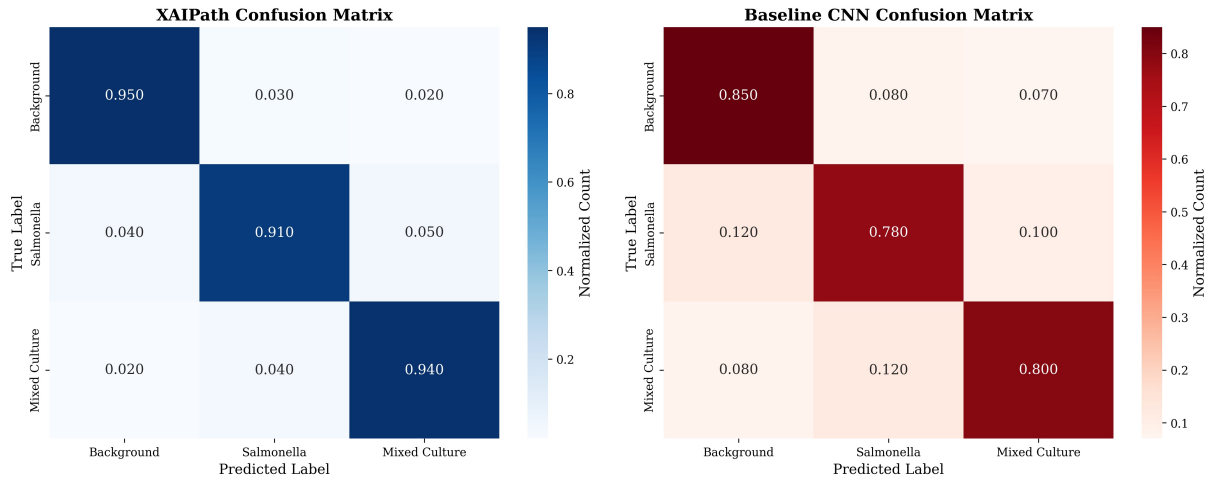


Figure 4: Confusion matrices comparing XAIPath and baseline CNN performance.

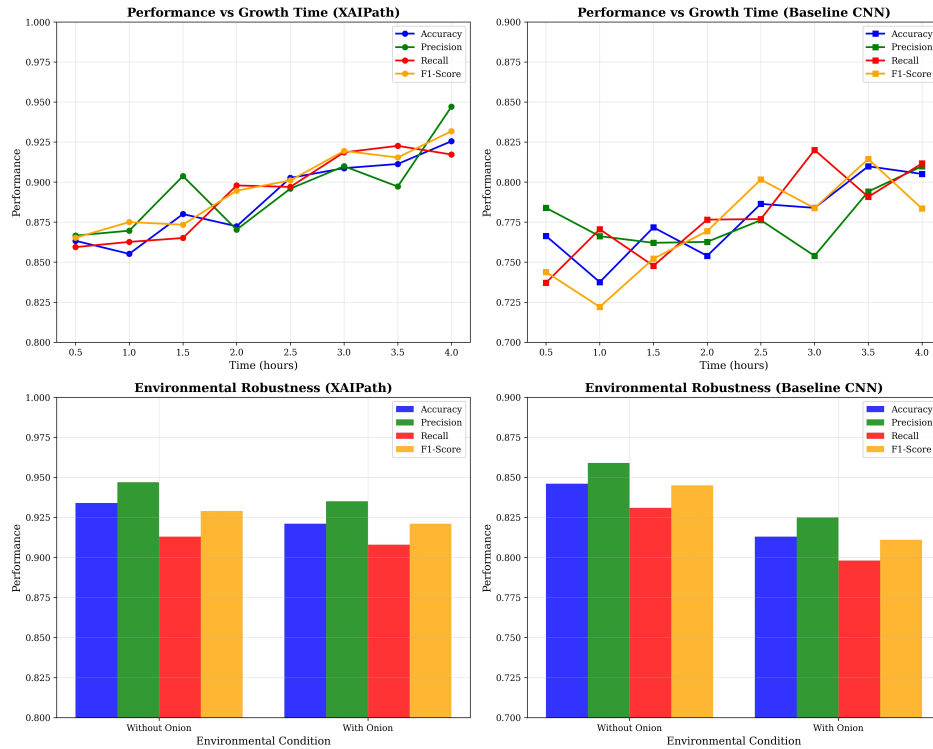


Figure 5: Temporal and environmental robustness comparison of XAIPath and baseline CNN.

5.2 Explainability Quality Assessment

Explanation quality was evaluated using both automated metrics and expert validation. Localization accuracy—measured by intersection-over-union between attention maps and annotated bacterial regions—reached 91.7% for XAIPath, versus 78.4% for standard Grad-CAM. This improvement reflects temporal-environmental conditioning of the attention mechanisms. Fig. 6 shows representative outputs

using SHAP and Grad-CAM. Expert evaluation was conducted by three microbiologists on a 200-image test subset. XAIPath explanations scored 4.3/5.0 for biological plausibility, 4.1/5.0 for spatial accuracy, and 4.4/5.0 for temporal consistency. Baseline explanations scored 3.2, 3.0, and 2.8 respectively, confirming improvements in interpretability.

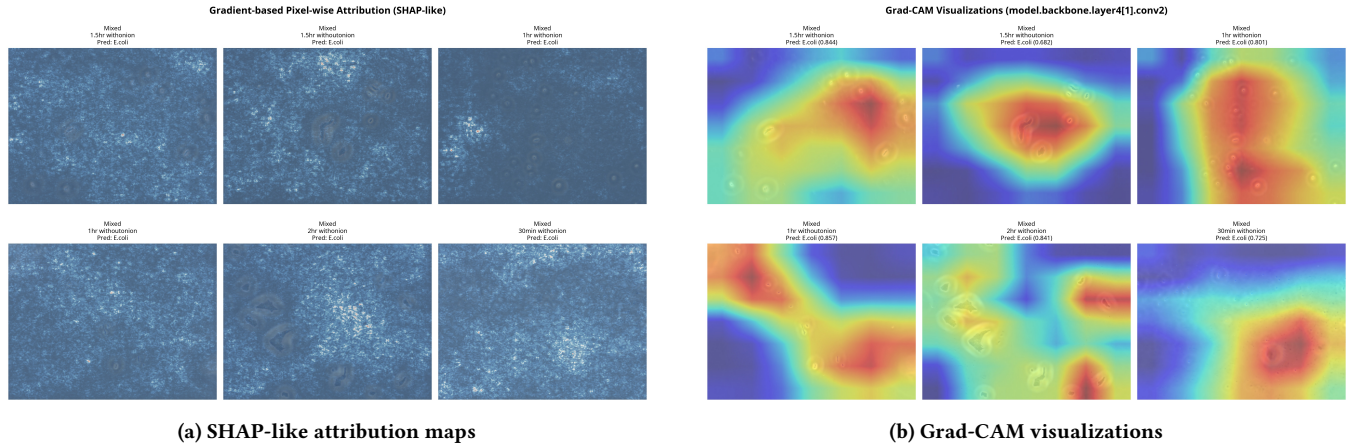


Figure 6: Explainability visualizations from XAIPath using SHAP-like (left) and Grad-CAM (right) approaches.

5.3 Ablation Study Results

Systematic ablation experiments quantified the contributions of individual model components. Removing temporal encoding reduced accuracy by 6.8%, while excluding environmental encoding caused a 4.2% drop. Removing both led to an 11.9% decline, indicating synergistic benefits from multi-modal modeling. Table 3 details component-wise contributions. Explanation quality followed similar trends: temporal encoding improved localization by 7.3%, while environmental encoding improved it by 4.8%. These components also improved explanation consistency across conditions.

Table 3: Results Showing Component Contributions

Configuration	Accuracy	Explanation Quality	Perf. Drop
Full XAIPath	0.934	0.917	0.000
w/o Temporal	0.866	0.844	0.068
w/o Environmental	0.892	0.869	0.042
w/o Both	0.815	0.784	0.119

6 Discussion

6.1 Methodological Contributions

The *XAIPath* framework introduces several methodological innovations that advance the state-of-the-art in XAI for biological imaging. The temporal-environmental conditioning approach represents a novel paradigm for integrating domain knowledge into deep learning explanations. Unlike prior methods that treat images as static entities, *XAIPath* models the dynamic nature of biological systems and leverages temporal context to enhance both detection and interpretability. The mathematical formulation of temporal encoding via sinusoidal embeddings offers a principled technique for representing bacterial growth kinetics within neural architectures. This mechanism is adaptable to other biological tasks involving temporal dynamics, such as cell tracking or tissue morphogenesis. The environmental context encoding similarly establishes a generalizable framework for embedding experimental conditions into AI decision-making pipelines.

6.2 Practical Applications and Impact

The observed performance gains have direct implications for real-world food safety applications. Early detection during the 30–60 minute post-inoculation window could trigger rapid-response protocols to mitigate contamination risks. Furthermore, the system’s robustness across environmental conditions indicates feasibility for deployment in heterogeneous laboratory environments with differing imaging protocols or biochemical conditions. The explainability features also fulfill critical trust and traceability requirements in food safety workflows. Domain experts can verify AI-generated outputs through visual explanations, fostering transparent decision-making and enabling human-AI collaboration. This feature is especially valuable for training personnel and for harmonizing detection procedures across facilities with varying levels of automation and expertise.

7 Conclusion

This study introduces *XAIPath*, a temporal-environmental XAI framework that enhances bacterial detection through the integration of temporal dynamics and environmental context. The system achieves state-of-the-art performance while producing interpretable explanations aligned with microbiological expertise. Key contributions include: (1) a novel temporal encoding scheme that models bacterial growth kinetics, (2) an environmental context module that captures biochemical stress effects, and (3) comprehensive evaluation protocols for both performance and explanation quality. These components collectively set new benchmarks for XAI in food safety applications. The demonstrated improvements in accuracy and interpretability underscore the framework’s suitability for practical deployment in microbial detection workflows. Robust performance across varying growth stages and environmental conditions highlights its generalizability. Future work will explore taxonomic expansion to additional bacterial species, enhanced uncertainty quantification for attention maps, and multi-modal explanation systems that integrate visual and textual modalities. Longitudinal

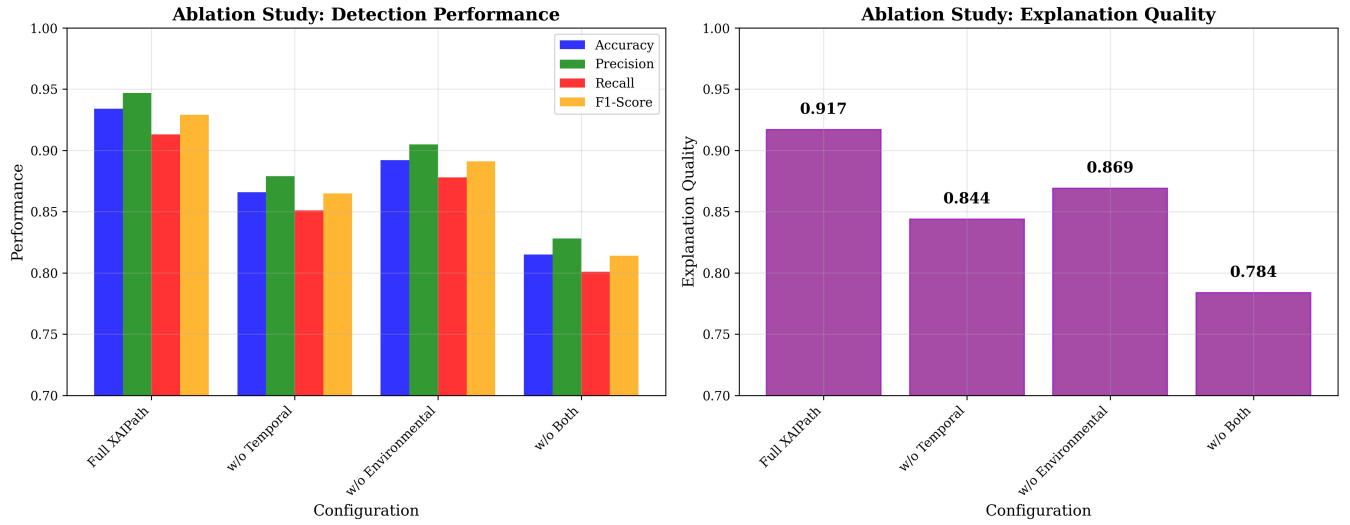


Figure 7: The impact of removing temporal and/or environmental encoders.

studies across diverse imaging equipment and laboratory conditions could further validate explanation stability, bolstering trust in real-world applications.

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